Reproducing and Extending Stochastic Switching in a Probabilistic Bandit Task Using Stickiness-Based Decision Models

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Adaptive decision-making in uncertain environments requires integrating reward history to guide flexible action selection. Beron et al. (2022) [1] showed that mice performing a probabilistic two-choice bandit task exhibit stochastic, history-dependent switching behaviour that is accurately captured by reinforcement learning and recursive logistic regression models with choice history bias. Here, we reproduce the key behavioural and generative modelling results under the original 80-20 reward condition and extend the analysis to a noisier 70-30 environment. We compare the behaviour of the recursive logistic regression (RFLR) model and hidden Markov model (HMM) using reward-based performance metrics, including reward rate and reward-per-switch efficiency. We introduce nonlinear stickiness mechanisms by implementing temporally gated and surprise-dependent choice history bias and assess whether dynamic stickiness improves model fit and switching efficiency. Model performance is assessed using likelihood-based and behavioural metrics. Our results demonstrate how adaptive stickiness mechanisms influence switching efficiency and behavioural flexibility and provide further insight into the computational principles underlying decision-making in uncertain environments.

INTRODUCTION

Behavioural flexibility, the capacity to adjust behaviour in response to changing environmental contingencies, is a fundamental component of adaptive decision making. This capacity is essential in natural environments, where reward availability is often uncertain and subject to change. Impairments in behavioural flexibility are associated with several neuropsychiatric disorders, highlighting the importance of understanding the computational mechanisms that support adaptive decision making (Izquierdo et al., 2017) [2]. Probabilistic reversal learning (PRL) tasks provide a controlled framework for studying behavioural flexibility. In these tasks, reward contingencies reverse unpredictably without explicit cues, requiring agents to infer environmental changes from noisy reward feedback. Because individual outcomes are probabilistic, optimal behaviour depends on integrating information across multiple trials rather than relying on deterministic win-stay/lose-switch strategies. Computational accounts of reversal learning have broadly focused on two classes of models: reinforcement learning and inference-based approaches (Daw et al., 2005[3]; Wilson et al., 2014[4]). Reinforcement learning models update action values incrementally based on reward prediction errors, whereas inference-based models estimate latent environmental states governing reward contingencies. These approaches make distinct predictions about behavioural adaptation and switching dynamics. Beron et al. (2022)[1] investigated these mechanisms in mice performing a dynamic two-choice bandit task. They found that switching behaviour was graded, stochastic, and strongly dependent on recent reward history. Importantly, behaviour was accurately captured by reinforcement learning and recursive logistic regression models incorporating stochastic action selection and choice history bias. These findings suggest that adaptive decision making may be governed by history-dependent evidence integration rather than deterministic switching rules. This project reproduces the principal behavioural and computational findings of Beron et al. (2022)[1], with particular emphasis on switching dynamics and adaptation to contingency reversals. We further extend this framework by examining model behaviour in more uncertain reward environments and introducing nonlinear stickiness mechanisms that dynamically modulate choice history bias. These extensions aim to provide deeper insight into the computational principles underlying stochastic action switching and behavioural flexibility. Critically, the present work extends the original study by explicitly testing whether stickiness is dynamically regulated across environments with different uncertainty levels and by implementing nonlinear stickiness mechanisms that allow adaptive modulation of behavioural persistence. Unlike the original study, which assumed fixed stickiness parameters, the present work explicitly tests whether stickiness can be dynamically modulated based on recent experience. By implementing surprise-dependent and temporally gated stickiness mechanisms, this study evaluates whether adaptive regulation of behavioural persistence provides functional advantages under uncertainty. This extension therefore moves beyond descriptive modelling to examine stickiness as a dynamically controlled computational process. We hypothesised that models incorporating adaptive stickiness mechanisms would better capture switching behaviour and maintain behavioural efficiency under increased environmental uncertainty. We further predicted that increasing environmental stochasticity would disproportionately impair inference-based models relative to stickiness-based models. Finally, we implement a multi-level gating framework that formally

compares distinct computational hypotheses regarding how surprise modulates behavioural persistence. Moving beyond descriptive model comparison, this phase introduces cross-validated likelihood optimisation, nested temporal parameter search, and generative validation to evaluate whether surprise-dependent gating operates at the level of policy drive, belief updating, or value learning. This framework enables a principled statistical comparison of alternative neurocomputational mechanisms.

METHODS: REPRODUCTION OF BERON ET AL. (2022)

Behavioural signature targeted for reproduction

Beron et al. (2022) reported that mice performing a dynamic two-choice bandit task exhibit graded, stochastic switching rather than deterministic win–stay/lose–switch behaviour. The key behavioural signature is the conditional switching probability $P(\text{switch} \mid \text{history})$, which increases with accumulating negative evidence but remains bounded below one. Despite stochastic feedback, mice achieve near-maximal reward rates, indicating efficient adaptation in uncertain environments.

Reproduction workflow and data provenance

To reproduce the principal findings, we used the publicly available dataset and analysis/model code released by Beron et al. (2022). Reproduction consisted of running the authors' pipeline to regenerate the key behavioural summaries and model-comparison figures reported in the original study. All reproduced figures were generated by executing the provided scripts/notebooks with fixed random seeds where applicable, and outputs were cross-checked against the paper for qualitative agreement (e.g., graded policy structure, model ranking, and HMM overswitching).

Computational models considered in Beron et al. (2022)

Beron et al. (2022) compared multiple model classes that differ in how reward history and environmental uncertainty are represented. We summarise these model families here to define the computational hypotheses evaluated in both the reproduction and extension analyses.

Win–Stay/Lose–Switch (WSLS). A heuristic strategy that repeats rewarded actions and switches after unrewarded outcomes. WSLS predicts near-deterministic switching after losses and therefore fails to capture graded behavioural responses.

Reinforcement learning (RL). A model-free approach that incrementally updates action values via reward prediction errors, typically paired with stochastic action selection. RL captures gradual adaptation but does not explicitly represent latent states of the environment.

Inference-based models (HMM). A latent-state model that infers the current reward contingency from past outcomes using Bayesian belief updating. Inference models can adapt rapidly following reversals but may over-switch under stochastic feedback if losses are over-interpreted as evidence for state change. Beliefs are updated according to:

$$P(s_t \mid o_{1:t}) \propto P(o_t \mid s_t) \sum_{s_{t-1}} P(s_t \mid s_{t-1}) P(s_{t-1} \mid o_{1:t-1}),$$

where $P(s_t \mid o_{1:t})$ is the posterior belief over latent reward states given observations $o_{1:t}$.

Recursive logistic regression (RFLR). A reduced model that predicts choices from accumulated choice–outcome history while incorporating a choice-history bias (stickiness). In the recursive formulation, the probability of choosing action a_t is:

$$P(a_t = 1) = \frac{1}{1 + \exp\left[-(\beta x_t + \alpha a_{t-1})\right]},$$

where x_t represents accumulated reward evidence, β is an inverse temperature controlling sensitivity to evidence, and α is the stickiness parameter biasing repetition of the previous choice.

Additional analysis scripts were used to regenerate behavioural summary metrics reported in the original study, including conditional switching probabilities, event-triggered switching dynamics, and quantitative comparisons between model and mouse policies. These analyses reproduced the key behavioural signatures described by Beron et al. (2022).

METHODS: EXTENSIONS

Behaviour under increased uncertainty (70–30 condition)

Generative model simulations. Using cross-validated parameter estimates, we generated synthetic behaviour in both reward environments (80-20 and 70-30) under three model classes: RFLR, Ideal HMM (Thompson sampling), and Sticky HMM. To ensure identical reversal structure across models, simulations were run on the empirical latent state sequences extracted from the experimental sessions. Reward delivery probabilities were set to 0.8/0.2 (80–20) or 0.7/0.3 (70–30). Transition probabilities were set to the task transition parameter used in the generative simulations (p_{switch}), consistent

with the probabilistic bandit structure. For each environment, RFLR parameters (α, β, τ) estimated via cross-validation were used to simulate behaviour. To assess parameter generalisation across environments, we additionally simulated the 70-30 condition using parameters fitted in the 80-20 environment.

Cross-validated parameter estimation across reward environments. To quantify how behavioural parameters depend on environmental uncertainty, the RFLR model was fit separately in the 80–20 and 70–30 reward environments. Parameter estimation was performed using 5-fold cross-validation at the session level to ensure robust generalisation and prevent information leakage between training and evaluation data. Sessions were randomly partitioned into five folds using a fixed random seed. For each fold, model parameters (α, β, τ) were fit on four training folds by maximising the log-likelihood of the observed choice sequence under the RFLR model. Optimisation was implemented using stochastic gradient descent (10,000 iterations; learning rate = 0.01). The fitted parameters were then evaluated on the held-out fold to assess out-of-sample predictive performance. This procedure was repeated for all folds, and the final parameter estimate for each environment was taken as the mean of parameter estimates across folds.

Behavioural metrics and model comparison. Model behaviour was evaluated using summary statistics computed at the session level. For each simulated session, we computed the total reward obtained per session and the total number of switches per session. These metrics were used to characterise the reward-cognitive trade-off across models. To quantify switching dynamics around environmental reversals, we computed the probability of switching as a function of trial distance relative to block transitions. Switch probabilities were aligned to empirical state transitions and averaged across sessions to characterise cognitive lag and adaptation dynamics. Finally, parameter estimates (α, β, τ) were compared across the 80-20 and 70-30 environments to assess systematic shifts associated with increased reward uncertainty.

Nonlinear stickiness mechanisms

To extend the fixed-stickiness framework, we implemented dynamic modulation of the stickiness parameter (α) based on recent environmental feedback. These mechanisms were applied to the RFLR, HMM, and forgetting Q-learning (F-Q) architectures and evaluated using generative simulations.

Informational gating: signed prediction-error modulation (RFLR and HMM). For the RFLR and HMM models, stickiness was dynamically scaled as a function of recent signed prediction error (PE). On each trial $t > 0$, effective stickiness was computed as

Multi-level gating framework and optimal evalusation

Model hypotheses. Phase 4 evaluates distinct computational hypotheses for how agents use environmental "surprise" to regulate choice persistence in the 2ABT. We compare (i) a direct policy model where surprise gates motor persistence trial-by-trial (dynamic RFLR), (ii) a belief-based model where Bayesian state inference is combined with motor-level stickiness (Sticky HMM), and (iii) a temporal/value baseline where switching behaviour emerges from windowed value dynamics (FQ value-gated). In addition, belief-reset variants test whether omission-driven surprise triggers change-point detection and belief reinitialisation.

Cross-validated optimisation. To prevent overfitting, parameters were fit using 5-fold cross-validation over sessions. For each fold, model parameters were optimised on the training set and evaluated on held-out sessions using mean log-likelihood (Mean LL) as the primary performance metric.

Nested search over temporal parameters. Discrete temporal parameters (e.g., commitment window N) were selected via a nested search procedure. The outer loop enumerated candidate N values, and the inner loop performed continuous optimisation of model parameters (e.g., α_{base} and surprise sensitivity γ) using L-BFGS-B with bounds. Parameter stability was assessed via across-fold variability, and the optimal N was selected via plurality vote across folds.

Model design: temporal gating and surprise-dependent modulation. To extend the computational framework beyond fixed stickiness, we implemented nonlinear stickiness mechanisms that dynamically regulate behavioural persistence. These extensions were applied to RFLR, HMM, and forgetting Q-learning (F-Q) models. Temporal gating was implemented as a policy-level temporal lock applied after predicted switches. Specifically, when the model's policy predicted a switch in choice, the selected action was locked and repeated for the subsequent N trials, preventing immediate reversal. This mechanism operates at the level of policy output probabilities and produces locked choice and switch sequences used for behavioural analysis. Informational gating was implemented via surprise-dependent modulation of stickiness, where effective stickiness varies with recent prediction error:

$$\alpha_{\text{dynamic}} = \alpha_{\text{base}} e^{-\gamma E},$$

with E denoting accumulated prediction error over the previous N trials and γ controlling sensitivity to surprise. This formulation decreases stickiness when outcomes are surprising and increases persistence when predictions are reliable.

Simulation protocol and recorded outputs. Synthetic behavioural sessions were generated using the

same probabilistic bandit task structure described above. Models were simulated across multiple sessions, and we recorded the dynamic stickiness trace, switching frequency, and (where applicable) value estimates (e.g., Q-values for F-Q). These outputs were used to assess how adaptive stickiness regulates the balance between persistence and flexibility under stochastic feedback.

Implementation details. All extension simulations were implemented in Python using custom code written for this project. For nonlinear stickiness experiments, synthetic behavioural data were generated for 5 independent sessions of 120 trials each, with reward contingencies evolving according to a two-state Markov process with transition probability $P(\text{switch}) = 0.1$. Behavioural metrics were averaged across sessions. Dynamic stickiness was modulated according to accumulated prediction error over N-step windows, with parameter values matched to those used in the computational implementation. Temporal gating was implemented by enforcing a fixed choice for N trials following a switch, while informational gating was implemented by updating the stickiness parameter on each trial according to the accumulated prediction error window. We hypothesised that dynamic stickiness would improve switching efficiency by preventing excessive switching in response to stochastic reward fluctuations and, where evaluated against baseline models, improve alignment with mouse switching dynamics.

Reproduction: quantitative agreement for all models. *Model-predicted vs mouse-observed* $P(\text{switch} \mid \text{history})$ *across histories (dotted identity line).*

RESULTS

Reproduction of Beron et al. (2022)

We first evaluated whether the implemented models reproduced the key behavioural signatures reported by Beron et al. Conditional switching probabilities computed directly from mouse behaviour exhibited graded dependence on recent reward history and remained bounded below one (Fig. 1), confirming that switching behaviour is stochastic rather than deterministic. Consistent with Beron et al. (2022), our reproduced model comparisons showed clear differences in their ability to capture behavioural dynamics. Logistic regression, RFLR, and forgetting Q-learning reproduced the gradual adaptation in both choice probability and switching behaviour around contingency reversals (Fig. 2). In contrast, the HMM exhibited sharper transitions and elevated switching rates relative to mice. These differences were also evident in the structure of conditional switching policies (Fig. 3). The RFLR closely matched the graded switching probabilities observed in mouse behaviour, whereas the HMM overestimated switching probability across many histories. Quantitative comparison confirmed this pattern: RFLR achieved the highest agreement with mouse switching probabilities (correlation = 0.93, SSE = 0.009), followed by logistic regression and forgetting Q-learning, while the HMM performed substantially worse (correlation = 0.72, SSE = 0.054; Table II). Scatter plots comparing model-predicted and mouse switching probabilities further illustrated the close correspondence for RFLR and larger deviations for the HMM (Figs. 4–5). Together, these results successfully reproduce the main findings of Beron et al., demonstrating that mouse behaviour is best explained by models incorporating stochastic evidence integration and choice history bias.

Validation of reproduction

These results confirm that our implementation accurately reproduces the key behavioural and modelling findings of Beron et al. (2022), validating the simulation framework for further extensions.

Extension: Behaviour under increased uncertainty (70–30 condition)

Modulation of parameters across environments: Cross-validated parameter estimates revealed systematic differences between the 80–20 and 70–30 reward environments (Fig. 6). In the 70–30 condition, the stickiness parameter (α) was higher than in the 80–20 condition, indicating greater behavioural persistence under increased reward stochasticity. By contrast, the inverse temperature parameter (β) and integration timescale (τ) exhibited comparatively smaller changes across environments. Among the three fitted parameters, α showed the largest shift between environments (Fig. 6), consistent with the interpretation that behavioural persistence is particularly sensitive to reward uncertainty. These results indicate that adaptation to increased environmental noise is primarily reflected in changes to stickiness rather than substantial alterations in reward sensitivity or evidence integration dynamics.

Ideal HMM exhibits elevated switching under increased uncertainty: Model performance in the 70–30 condition is summarised in Table I and visualised in Fig. 7. Although the Ideal HMM achieved marginally higher total reward per session than the RFLR, it did so at the cost of substantially elevated switching frequency. The HMM produced markedly more switches per session than both the mice and the RFLR model, occupying a high-switch regime in the reward-switch trade-off space (Fig. 7). In the 70-30 condition, stochastic losses frequently occur even when selecting the correct option. The inference-based HMM interprets these losses as potential state changes, leading to rapid belief updating

and increased switching. While this strategy maintains high overall reward, it does so at a substantially greater behavioural cost. By contrast, the RFLR model maintains switching rates comparable to those observed in mouse behaviour while achieving similar reward levels, indicating a more conservative trade-off between reward acquisition and behavioural flexibility.

Cognitive lag under noise: Event-triggered switching dynamics were quantified by aligning the trial-to-trial switch indicator (δchoice|) to empirical contingency reversals and averaging across sessions (Fig. 8). In the 70–30 condition, mouse switching probability increases gradually following reversals ($t = 0$), exhibiting a delayed and bounded rise rather than an abrupt transition in behaviour. Generative simulations of the RFLR model evaluated on identical state sequences reveal a clear dependence on environment-specific parametrisation. When fitted directly in the 70–30 condition, the RFLR model closely reproduces the empirical time course, capturing both the baseline switching probability and the gradual post-reversal increase. In contrast, applying parameters fitted in the 80–20 environment to the 70–30 condition produces elevated baseline switching and exxagerated post-reversal responses, resulting in systematic overshoot relative to the mouse data. This discrepancy is quantified in the lower panel of Fig. 8, which plots the deviation from empirical behaviour normalised by the standard error of the mouse switching probability at each time point,

$$\bar{PE}_t = \frac{1}{|\mathcal{B}_t|} \sum_{PE \in \mathcal{B}_t} PE$$

denotes the mean signed prediction error stored in a finite sliding buffer $\mathcal{B}_t$ of maximum length N. In implementation, this buffer was realised as a deque structure updated on each trial. During the initial trials of a session, the buffer contains fewer than N elements, and the mean is computed over the available prediction errors. On the first trial, $\alpha_0 = \alpha_{\text{base}}$, and the initial prediction error is computed relative to a neutral reward expectation.

Because prediction errors are signed, this formulation produces asymmetric multiplicative modulation of behavioural persistence. Sustained negative prediction errors (e.g., unexpected reward omissions) reduce α_t, weakening choice repetition and facilitating switching, whereas sustained positive prediction errors increase α_t, reinforcing recent choices. Using the mean (rather than the sum) ensures that the magnitude of modulation remains comparable across window sizes.

Prediction errors were defined according to each model's internal expectations: for RFLR, prediction error was computed relative to the reward expectation derived from its evidence accumulator; for HMM models, prediction error was computed relative to the belief-weighted expected reward under the current posterior.

Belief reset under surprise (HMM variants). For HMM extensions incorporating reset dynamics, posterior beliefs were additionally mixed with a uniform prior under conditions of sustained surprise. Specifically, after standard Bayesian updating produced a posterior π_t^{raw}, the belief state was adjusted as

$$\pi_t = (1 - s_t)\pi_t^{\text{raw}} + s_t \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix},$$

where the surprise signal s_t was derived from recent prediction error. Two variants were implemented: (i) an asymmetric reset in which only sustained negative prediction errors (reward omissions) increased s_t, and (ii) a symmetric reset in which s_t scaled with the magnitude of prediction error $|\bar{PE}_t|$. This soft reset increases uncertainty following sustained unexpected outcomes while preserving graded belief dynamics.

Value-based temporal gating (F-Q). Two extensions of the forgetting Q-learning architecture were implemented.

In the value-gated variant, stickiness α remained constant and learning was temporally discretised at the value level. Q-values were updated only after accumulation of outcomes over fixed windows of length N. Within each window, Q-values remained fixed and were updated using the mean reward obtained for each action during that window.

In the nonlinear stickiness F-Q variant, stickiness was additionally modulated according to the mean absolute reward prediction error:

Unlike the RFLR and HMM implementations, this formulation is symmetric with respect to error sign and scales persistence according to surprise magnitude.

Simulation protocol. All nonlinear variants were evaluated using generative simulations under the same probabilistic bandit structure described above. Synthetic sessions consisted of 120 trials with state transitions governed by a two-state Markov process ($P(\text{switch}) = 0.1$). For each session, we recorded dynamic stickiness traces (α_t), switching frequency, and (where applicable) evolving Q-values. Behavioural summaries were computed at the session level and averaged across simulations. For each nonlinear model, switching and reward metrics were compared directly against corresponding fixed-stickiness baselines under identical generative conditions. These nonlinear variants were evaluated exclusively using generative simulations and were not refit to empirical data.

Hyperparameter optimisation. Window length N and modulation sensitivity γ were selected using nested grid search to maximise cross-validated log-likelihood. For each candidate pair (N, γ), models were evaluated in fitting mode, and optimal hyperparameters were chosen based on out-of-sample predictive performance.

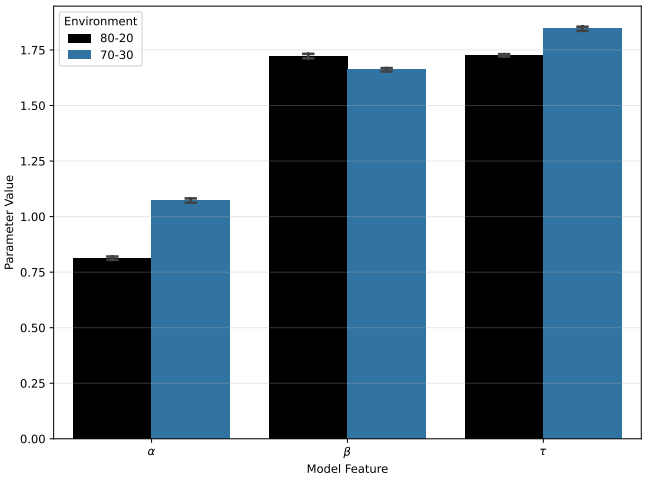

FIG. 1: **Context-dependent modulation of model parameters across reward environments.** Estimated parameter values for stickiness (α), inverse temperature (β), and integration timescale (τ) in the 80–20 and 70–30 reward conditions. Increasing environmental uncertainty was associated with a marked increase in effective stickiness (α) and integration timescale (τ), indicating greater behavioural persistence and slower evidence integration under noisy reward feedback. In contrast, inverse temperature (β) showed only modest variation across conditions. Error bars denote standard error across sessions. These results demonstrate that behavioural parameters adapt systematically to environmental uncertainty, supporting the hypothesis that stickiness is dynamically regulated to stabilise behaviour under stochastic reward conditions.

TABLE I: **70–30 extension: model performance comparison.** Summary statistics comparing mouse behaviour, RFLR, and HMM in the 70–30 reward condition. Reward rate quantifies overall performance, switches/session measures switching frequency, and reward-per-switch reflects switching efficiency. The HMM achieves slightly higher reward rate but switches excessively, resulting in substantially lower switching efficiency.

Model	Reward/session	Reward/switch	Switches/session
Mouse	0.595	11.960	51.6
RFLR	0.595	8.546	55.9
HMM	0.612	2.652	181.6

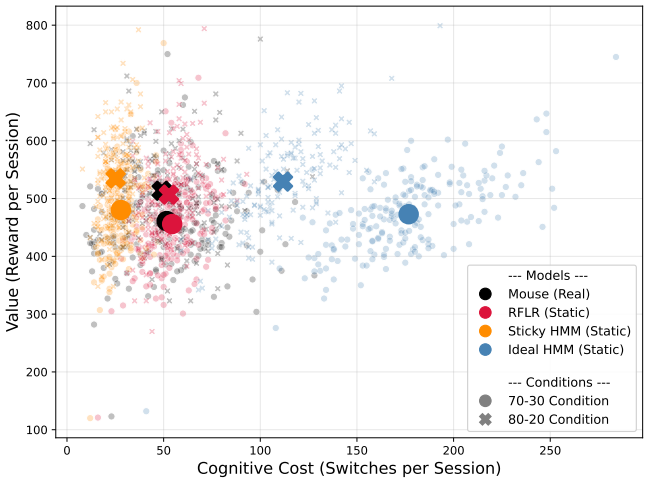

FIG. 2: **Reward–switch trade-off across environments.** Total reward per session is plotted against switching frequency (switches per session). Circles denote the 70–30 condition and crosses denote the 80–20 condition. Small translucent markers represent individual sessions, while larger opaque markers indicate model or animal means. Under increased uncertainty (70–30), empirical mice (black) maintain relatively low switching rates despite a modest reduction in total reward. In contrast, the Ideal HMM (blue) achieves comparable or slightly higher reward but at a substantially elevated switching cost. The absence of stickiness leads the inference-based model to interpret stochastic losses as state transitions, producing excessive switching not observed in mouse behaviour. The RFLR model remains closer to the empirical reward–switch regime, reflecting a more balanced trade-off between flexibility and persistence.

TABLE II: Model comparison summary (reproduction).

Model	Correlation (cprobs)	SSE
Logistic regression	0.91	0.012
RFLR	0.93	0.009
HMM	0.72	0.054
Forgetting Q-learning	0.90	0.015

RESULTS

Reproduction of Beron et al. (2022)

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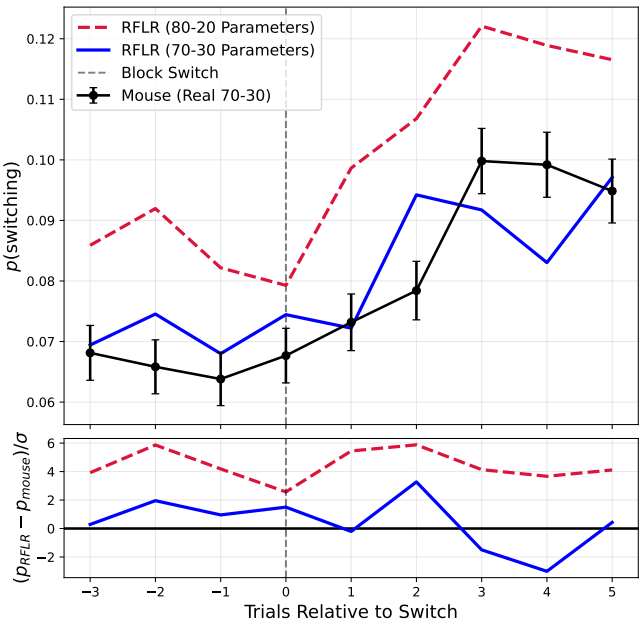

FIG. 3: **Cross-environment validation of switching dynamics and environment-specific stickiness. Upper panel:** Empirical switching probability p(switch) in the 70–30 condition aligned to contingency reversals (circles denote mean across sessions; error bars represent the standard error of the mean). Solid and dashed curves show generative simulations of the RFLR model evaluated on identical state sequences. When parameterised using 80–20 fitted parameters (crimson dashed), the model exhibits elevated baseline switching and exaggerated post-reversal responses relative to the empirical data. In contrast, the RFLR model refitted to the 70–30 environment (blue solid) closely reproduces the gradual, bounded increase in switching probability observed in mice. **Lower panel:** Normalised deviation from mouse behaviour, computed as $(p_{\text{model}} - p_{\text{mouse}})/\sigma_{\text{mouse}}$, where σ_{mouse} denotes the standard error of the empirical switching probability at each time point. The horizontal black line at zero represents the empirical mean. Deviations remain near zero for the 70–30–fitted model, whereas the 80–20 parameterisation shows systematic positive residuals following reversals, indicating oversensitive switching under increased uncertainty. These results demonstrate that stickiness parameters calibrated in one reward environment do not generalise across environments with different volatility.

Consistent with Beron et al. (2022), our reproduced model comparisons showed clear differences in their ability to capture behavioural dynamics. Logistic regression, RFLR, and forgetting Q-learning reproduced the gradual adaptation in both choice probability and switching behaviour around contingency reversals (Fig. 2). In contrast, the HMM exhibited sharper transitions and elevated switching rates relative to mice. These differences were also evident in the structure of conditional switching policies (Fig. 3). The RFLR closely matched the graded switching probabilities observed in mouse behaviour, whereas the HMM overestimated switching probability across many histories. Quantitative comparison confirmed this pattern: RFLR achieved the highest agreement with mouse switching probabilities (correlation = 0.93, SSE = 0.009), followed by logistic regression and forgetting Q-learning, while the HMM performed substantially worse (correlation = 0.72, SSE = 0.054; Table II). Scatter plots comparing model-predicted and mouse switching probabilities further illustrated the close correspondence for RFLR and larger deviations for the HMM (Figs. 4–5). Together, these results successfully reproduce the main findings of Beron et al., demonstrating that mouse behaviour is best explained by models incorporating stochastic evidence integration and choice history bias.

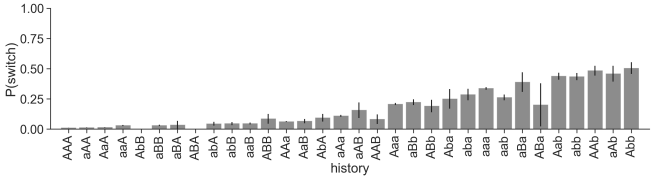

FIG. 4: **Nonparametric switching policy (mouse baseline).** Empirical conditional switching probabilities P(switch | history) for history length 3 computed directly from mouse behaviour (no model fit). Switch probability varies systematically with recent action–outcome history and remains bounded well below 1, consistent with graded and stochastic switching as reported by Beron et al and reproduced from our implementation of the dataset.

Validation of reproduction

These results confirm that our implementation accurately reproduces the key behavioural and modelling findings of Beron et al. (2022), validating the simulation framework for further extensions.

Extension: Behaviour under increased uncertainty (70–30 condition)

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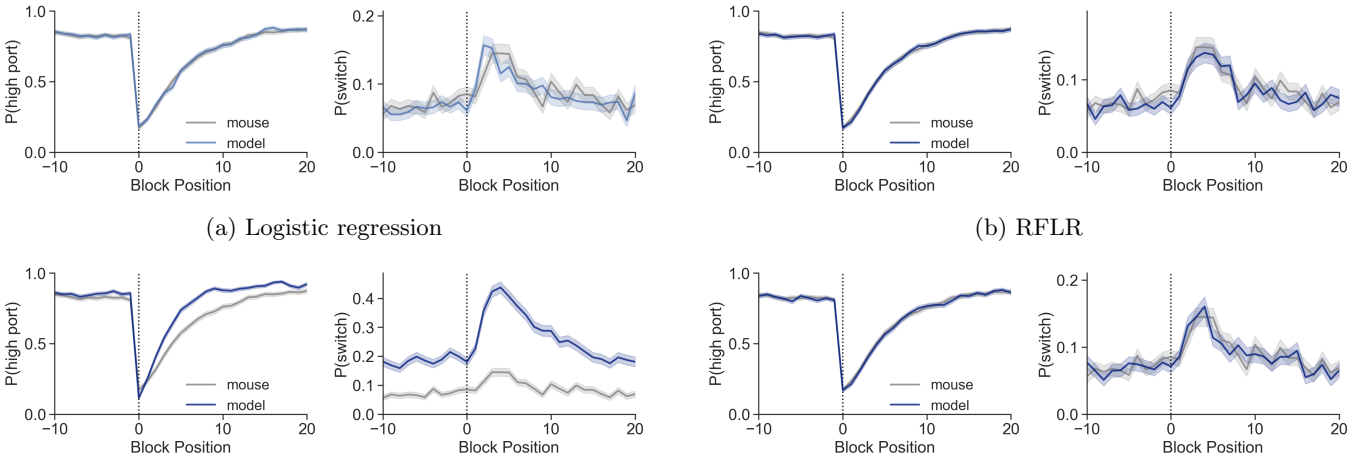
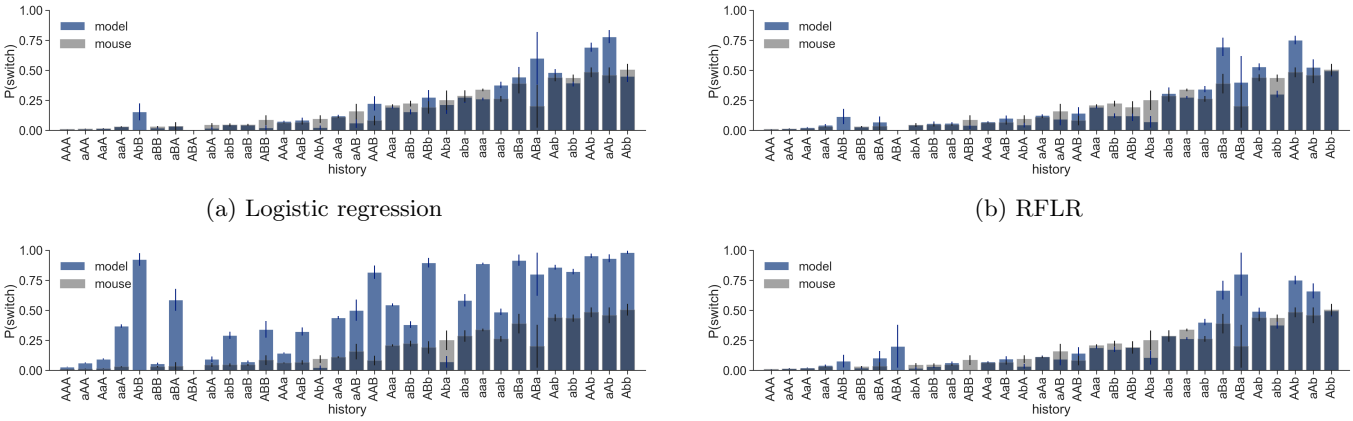


(a) Logistic regression (b) RFLR

(c) HMM (d) Forgetting Q-learning

FIG. 5: **Reproduction: behaviour around contingency reversals.** Model (blue) vs mouse (grey) for P(high port) (left) and P(switch) (right) as a function of block position (dotted line at 0 denotes reversal). This reproduces the block-transition evaluation used in Beron et al. and enables qualitative comparison of how each model captures adaptation following reversals.



(a) Logistic regression (b) RFLR

(c) HMM (d) Forgetting Q-learning

FIG. 6: **Reproduction: conditional switching by recent history.** Mouse (grey) and model (blue) conditional switch probabilities P(switch | history) for history length 3 (histories sorted by mouse P(switch)). This reproduces the core "policy-shape" comparison in Beron et al. and highlights model-specific deviations from the empirical switching structure.

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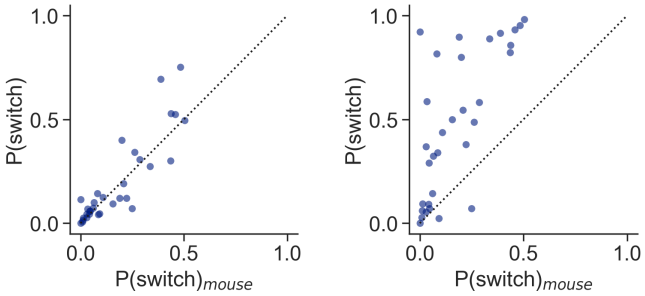
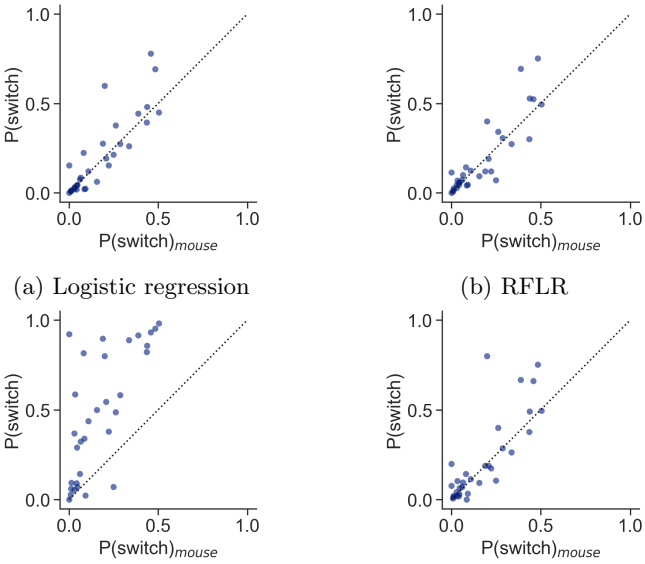


(a) RFLR (b) HMM

FIG. 7: **Reproduction: quantitative fit to conditional switching.** Scatter plots compare model-predicted vs mouse-observed $P(\text{switch} \mid \text{history})$ across histories (dotted line indicates identity). These plots provide a compact quantitative summary of agreement between empirical and model policies.



(a) Logistic regression (b) RFLR

(c) HMM (d) Forgetting Q-learning

behavioural cost. By contrast, the RFLR model maintains switching rates comparable to those observed in mouse behaviour while achieving similar reward levels, indicating a more conservative trade-off between reward acquisition and behavioural flexibility.

Cognitive lag under noise: Event-triggered switching dynamics were quantified by aligning the trial-to-trial switch indicator (δchoice|) to empirical contingency reversals and averaging across sessions (Fig. 8). In the 70–30 condition, mouse switching probability increases gradually following reversals ($t = 0$), exhibiting a delayed and bounded rise rather than an abrupt transition in behaviour. Generative simulations of the RFLR model evaluated on identical state sequences reveal a clear dependence on environment-specific parametrisation. When fitted directly in the 70–30 condition, the RFLR model closely reproduces the empirical time course, capturing both the baseline switching probability and the gradual post-reversal increase. In contrast, applying parameters fitted in the 80–20 environment to the 70–30 condition produces elevated baseline switching and exxagerated post-reversal responses, resulting in systematic overshoot relative to the mouse data. This discrepancy is quantified in the lower panel of Fig. 8, which plots the deviation from empirical behaviour normalised by the standard error of the mouse switching probability at each time point,

$$\frac{p_{\text{model}} - p_{\text{mouse}}}{\sigma_{\text{mouse}}}.$$

Deviations remain near zero for the 70–30 fitted model, whereas the 80–20 parameterisation shows consistent positive residuals following reversals, indicating oversensitive switching under increased uncertainty.

Static parameters are environment specific: Cross-environment simulations further demonstrate that parameters fitted in one reward environment do not yield equivalent behavioural trade-offs when applied to another. As illustrated in Fig. 7, the 70–30 environment shifts the reward–switch frontier, and parameter settings calibrated in the 80–20 condition fail to preserve switching dynamics under increased stochasticity. These results highlight that stickiness is environment-dependent and that static parameter settings do not generalise across levels of reward volatility.

Extension: Nonlinear stickiness mechanisms

We next examined how adaptive modulation of stickiness alters behavioural dynamics under stochastic feedback.

Dynamic modulation of persistence. Introducing prediction-error–dependent modulation produced substantial variation in effective stickiness over time. Although baseline stickiness was set to $\alpha_{\text{base}} = 1.2$, mean dynamic stickiness was computed by first averaging α_t across trials within each simulated session and subsequently averaging across sessions. Under nonlinear modulation, mean α_t was reduced to 0.351 in the RFLR model, 0.120 in the HMM model, and 0.105 in the F-Q model. These reductions reflect sustained periods of negative signed prediction error in the RFLR and HMM models, or elevated absolute reward prediction error in the nonlinear F-Q model, which multiplicatively downregulated behavioural persistence. In the RFLR and HMM models, modulation was asymmetric: sustained negative signed prediction errors weakened persistence, whereas sustained positive signed prediction errors reinforced recent choices. In contrast, the nonlinear F-Q implementation responded symmetrically to prediction

error magnitude, scaling persistence according to absolute surprise irrespective of sign.

Effects on switching dynamics. Adaptive modulation of stickiness altered switching behaviour without modifying the underlying reward-learning rules. Across sessions, dynamic stickiness reduced overall switching frequency relative to fixed-stickiness baselines. In particular, surprise-dependent down-regulation of α_t during periods of inconsistent feedback limited noise-driven switching while preserving persistence during stable reward phases. For HMM reset variants, soft belief mixing toward a uniform prior increased uncertainty following sustained surprise, preventing runaway overconfidence and moderating excessive switching.

Value stability under temporal gating. In the value-gated F-Q model, windowed Q-value updates produced more conservative and stable value estimates. Final Q-values remained partially separated (left = 0.394, right = 0.499), reflecting slower but more robust value differentiation under temporally discretised updating.

Behavioural efficiency. Comparison against corresponding fixed-stickiness baselines demonstrated that Nonlinear modulation reduced switches per session and increased reward-per-switch. Temporal discretisation and prediction-error gating reduced switches per session and increased reward-per-switch, with minimal impact on total reward. These findings indicate that adaptive regulation of stickiness is sufficient to improve behavioural efficiency under stochastic reward feedback.

Importantly, these results arise from generative simulations rather than formal model fitting. They therefore demonstrate that dynamic stickiness provides a plausible computational mechanism for regulating the trade-off between persistence and flexibility, without asserting that animals necessarily implement these exact mechanisms.

History-dependent switching structure. History-conditioned switching profiles reveal whether models reproduce the mouse micro-policy for responding to sequences of rewards and omissions. Across environments, dynamic gating models capture graded increases in switching probability following omission-rich histories, while differing in the extent to which they exaggerate or smooth switching after extended omission streaks.

Generative efficiency trade-off. In generative mode, models were evaluated on their ability to reproduce the mouse's reward–switching trade-off. The reward–cognitive cost frontier reveals whether a model achieves comparable reward while avoiding excessive switching. Dynamic gating models cluster closer to the mouse distribution than static baselines, indicating that improvements in predictive fit translate into autonomous behavioural efficiency rather than purely descriptive curve-fitting.

Across models, transition plots highlight differences in reversal responsiveness: belief-based models can react sharply to inferred state changes, whereas value-gated learning produces smoother, slower transitions consistent with temporally aggregated updates.

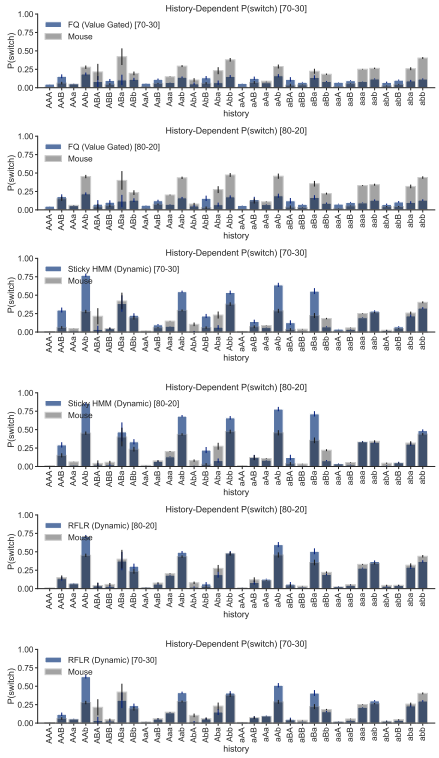
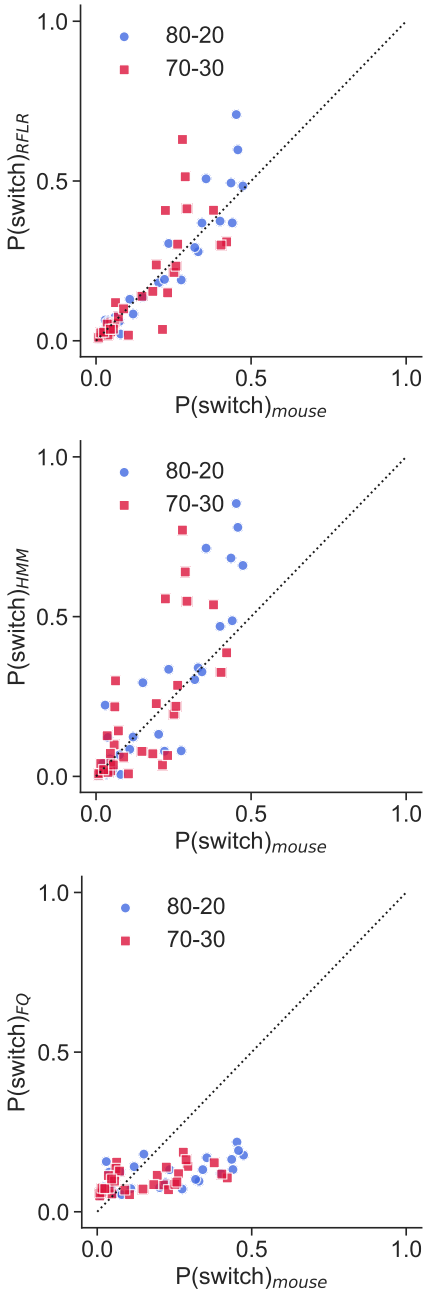


FIG. 9: **History-dependent switching under multi-level surprise gating.** Conditional switching probabilities $P(\text{switch} \mid \text{history})$ for value-gated FQ, Sticky HMM (dynamic), and dynamic RFLR models in the 70–30 and 80–20 reward environments. Grey bars denote empirical mouse behaviour and blue bars denote model predictions. This micro-dynamic analysis tests whether models capture trial-history-level switching structure beyond aggregate switch rates.

DISCUSSION

This study reproduced and extended the findings of Beron et al. (2022), providing further insight into the computational mechanisms underlying stochastic action switching. Consistent with the original study, mouse behaviour exhibited graded, history-dependent switching that was accurately captured by the recursive logistic regression (RFLR) model but less well described by the hidden Markov model (HMM). In particular, the RFLR more closely matched the structure of conditional switching probabilities and behavioural adaptation around contingency reversals, supporting the interpretation that mouse behaviour reflects stochastic evidence integration with choice history bias rather than optimal latent-state inference. Mechanistically, these re-



Qualitatively, dynamic RFLR exhibits the closest alignment to the identity line, whereas Sticky HMM shows systematic deviations consistent with oversensitivity to omission histories. The value-gated FQ model shows intermediate agreement, consistent with slower adaptation in trial-by-trial switching.

FIG. 10: **Predictive micro-fit: correlation of history-dependent switching.** Scatter plots compare empirical mouse $P(\text{switch}|\text{history})$ against model-predicted values across reward histories for the 80–20 and 70–30 environments. Points closer to the identity line indicate better trial-history-level agreement.

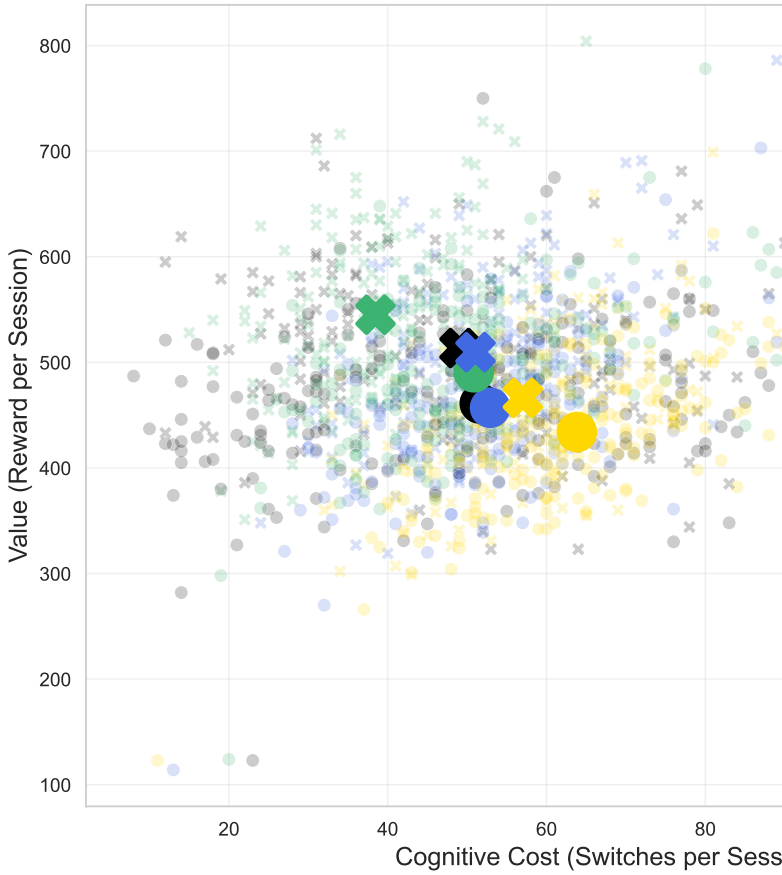


FIG. 11: **Generative validation: reward–switching efficiency frontier.** Generative simulations using best-fit parameters compare accumulated reward per session (value) against cognitive cost (switches per session) for mouse behaviour and dynamic gating models in both 80–20 and 70–30 environments. A good generative model should match the mouse's position in the reward–cost plane, indicating biologically plausible autonomous behaviour.

sults demonstrate that stickiness need not be fixed but can be dynamically regulated according to environmental feedback. Surprise-dependent modulation allows the agent to reduce persistence when recent outcomes contradict expectations, facilitating behavioural flexibility. Conversely, temporal gating prevents rapid, noise-driven switching by enforcing short-term commitment following behavioural changes. These mechanisms address complementary limitations of the original models. Fixed stickiness improves stability but can impair flexibility when environmental contingencies change. In contrast, adaptive stickiness enables flexible adjustment of persistence according to environmental volatility. These findings suggest that behavioural stickiness may function as a dynamically regulated control parameter rather than a fixed behavioural bias. This interpretation is consistent with

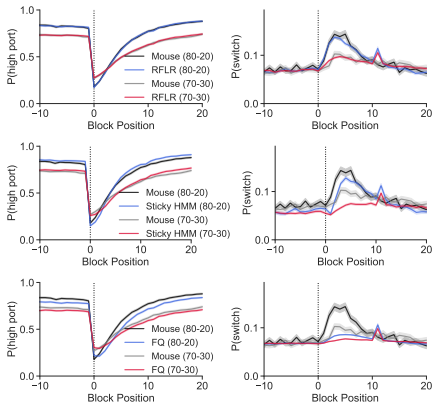


FIG. 12: **Transition dynamics around contingency changes.** Predictive-mode transition dynamics show how models track changes in latent reward state across block positions, comparing the probability of choosing the high port and the probability of switching in 80–20 and 70–30 environments. Models differ in responsiveness to reversals, revealing whether surprise gating acts primarily on motor persistence (RFLR/HMM) or via slower value updates (FQ).

the hypothesis that animals adjust their persistence in response to environmental uncertainty to optimise reward acquisition. However, several limitations should be acknowledged. First, simulations were performed using parameter values derived from previously published fits rather than independently refitting all models to the experimental dataset. Second, the nonlinear stickiness mechanisms were evaluated only using generative simulations rather than formal model fitting, limiting conclusions about whether animals truly implement such mechanisms. Finally, the relatively small number of simulated sessions may limit the statistical power of comparisons. Future work should perform full parameter optimisation and model comparison using larger datasets.

Phase 4 takeaway. Together, likelihood-based comparison, micro-dynamic switching structure, and generative efficiency provide a convergent evaluation of competing hypotheses. This multi-level framework tests not only whether a model can describe the observed choices, but also whether it reproduces the functional reward–switching trade-offs and reversal responsiveness characteristic of mouse behaviour under uncertainty.

CONCLUSION

This study successfully reproduced and extended the computational findings of Beron et al. (2022), providing further insight into the mechanisms underlying stochastic switching behaviour in uncertain environments. Consistent with the original study, mouse behavioural dynamics were accurately captured by the recursive logistic regression model, whereas the hidden Markov model exhibited excessive switching due to its sensitivity to stochastic reward outcomes. Extending the task to a more uncertain 70–30 reward environment demonstrated that stickiness mechanisms confer behavioural robustness, enabling more efficient reward accumulation by preventing excessive switching. Further extension using nonlinear stickiness mechanisms showed that adaptive modulation of stickiness allows models to dynamically balance persistence and flexibility. Surprise-dependent stickiness reduced behavioural persistence when recent outcomes were unreliable, while temporal gating stabilised behaviour by preventing rapid, noise-driven switching. Together, these findings demonstrate that stickiness plays a fundamental computational role in adaptive decision-making. Rather than acting as a fixed behavioural bias, stickiness may function as a dynamically regulated control parameter that enables efficient behaviour in stochastic environments. These results support the broader view that simple, history-dependent mechanisms can approximate optimal performance while maintaining robustness to environmental uncertainty. This evidence supports the hypothesis that adaptive stickiness improves behavioural efficiency under environmental uncertainty. More broadly, these findings demonstrate that adaptive regulation of behavioural persistence provides a simple yet powerful mechanism for achieving robust performance in uncertain environments. This suggests that dynamic stickiness may represent a fundamental computational principle underlying flexible decision-making.

* All figures were generated using custom Python code written by the authors, unless stated otherwise, e.g., for reproduction of original results. The full analysis and figure-generation pipeline is available at `https://github.com/BOBRICE1/Stochastic-Switching-Bandits`.

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